

SAMPLE QUESTION PAPER MATHEMATICS (311)

Time: 3 hrs

Maximum Marks: 100

Note:

- i. This question paper consists of 45 questions in all.
- ii. All questions are compulsory.
- iii. Marks are given against each question.
- iv. **Section A** consists of
 - a. **Q.No. 1 to 20** – Multiple Choice type questions (MCQs) carrying 1 mark each. Select and write the most appropriate option out of the four options given in each of these questions. An internal choice has been provided in some of these questions. You have to attempt only **one** of the given choices in such questions.
 - b. **Q.No. 21 to 29** – Objective type questions. **Q.No. 21 to 24** carry 02 marks each (with 2 sub-parts of 1 mark each), **Q.No. 25 to 28** carry 04 marks each (with 4 sub-parts of 1 mark each) and **Q.No. 29** carries 06 marks (with 6 sub-parts of 1 mark each). Attempt these questions as per the instructions given for each of the questions 18 – 29.
- v. **Section B** consists of
 - a. **Q.No. 30 to 38** – Very Short questions carrying 02 marks each.
 - b. **Q.No. 39 to 43** – Short Answer type questions carrying 04 marks each.
 - c. **Q.No. 44 to 45** – Long Answer type questions carrying 06 marks each.

SECTION A		
Q. No.	Questions	Marks
	<u>Q.No. 1 to 20</u> are the objective questions (MCQs) of 1 mark each: <i>An internal choice has been provided in some of these questions. You have to attempt only one of the given choices in such questions.</i>	
1.	(i) The coordinates of the midpoint of A (4, -1) and B(7, 2) are: a) $\left(\frac{3}{2}, \frac{1}{2}\right)$ b) $\left(\frac{3}{2}, \frac{3}{2}\right)$ c) $\left(\frac{11}{2}, \frac{3}{2}\right)$ d) $\left(\frac{11}{2}, \frac{1}{2}\right)$ OR (ii) The slope of the line segment joining the points A (2, 3) and B(6,-7) is: a) -2 b) 2 c) $-\frac{5}{2}$ d) $\frac{1}{2}$	1
2.	(i) The intercepts made by the line $3x - 2y + 12 = 0$ on the coordinate axis are : a) (4 and -6) b) (-4 and 6) c) (-4 and -6) d) (4 and 6) OR	1

	(ii) The equation of the line passing through (3, 7) and (-2, 5) is a) $2x - 5y = 29$ b) $2x - 5y + 29 = 0$ c) $2x + 5y + 29 = 0$ d) $2x + 5y = 29$	
3.	(i) The centre and radius of the circle $4x^2 + 4y^2 - 2x + 3y - 6 = 0$ are : a) $\left(\frac{1}{4}, \frac{3}{8}\right)$ and $\frac{\sqrt{109}}{8}$ b) $\left(-\frac{1}{4}, -\frac{3}{8}\right)$ and $\frac{\sqrt{109}}{8}$ c) $\left(\frac{1}{4}, -\frac{3}{8}\right)$ and $\frac{\sqrt{109}}{8}$ d) $\left(-\frac{1}{4}, \frac{3}{8}\right)$ and $\frac{\sqrt{109}}{8}$ OR (ii) The equation of circle whose centre is (3,4) and radius in 5. a) $x^2 + y^2 - 6x - 8y = 25$ b) $x^2 + y^2 + 6x + 8y = 50$ c) $x^2 + y^2 - 6x - 8y = 0$ d) $x^2 + y^2 + 6x + 8y = 0$	1
4.	The angles between the lines $2x+3y = 4$ and $3x-2y=7$ is a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{6}$	1
5.	The coordinates of the centroid of the triangle whose vertices are: $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) : a) $\left(\frac{x_1+y_1+x_3}{3}, \frac{y_1+x_2+x_3}{3}\right)$ b) $\left(\frac{x_1+x_2+x_3}{2}, \frac{y_1+y_2+y_3}{2}\right)$ c) $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ d) $\left(\frac{x_1+x_2+x_3}{2}, \frac{y_1+y_2+y_3}{3}\right)$	1

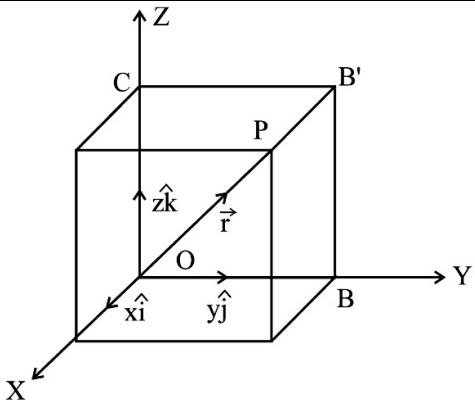
6.	(i) If $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, then $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, then the value of AB is: a) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ b) $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ d) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ OR (ii) If $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$, then A^n (where $n \in N$) is equal to: a) $\begin{pmatrix} 1 & na \\ 0 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 1 & n^2a \\ 0 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 1 & na \\ 0 & 0 \end{pmatrix}$ d) $\begin{pmatrix} n & na \\ 0 & n \end{pmatrix}$	1
7.	If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then the value of $A + A^T$ is: a) $\begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$ b) $\begin{pmatrix} 2 & 8 \\ 5 & 5 \end{pmatrix}$ c) $\begin{pmatrix} 2 & 5 \\ 8 & 5 \end{pmatrix}$ d) $\begin{pmatrix} 2 & -5 \\ 5 & 8 \end{pmatrix}$	1
8.	If $\begin{bmatrix} p+q & 2 \\ 5 & q \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 2 \end{bmatrix}$ then the value of p is: a) 6 b) 4	1

	c) 8 d) 5	
9.	If $f(x) = x^2$ and $g(x) = 3$, then the value of $f \circ g(x)$ is : a) 12 b) 15 c) 9 d) 18	1
10.	If $f(x) = x + 3$ for $x \in R$, then the value of $f^{-1}(x)$ is : a) $f^{-1}(x) = x - 3$ b) $f^{-1}(x) = x + 3$ c) $f^{-1}(x) = \frac{1}{x+3}$ d) $f^{-1}(x) = \frac{1}{x-3}$	1
11	Which of the following function from Z to itself are bijections? a) $f(x) = x^3$ b) $f(x) = x + 2$ c) $f(x) = 2x + 1$ d) $f(x) = x^3 + 1$	1
12.	If a binary operation $*$ is defined on the set Z of integers as $a*b = 3a-b$, then the value of $(2*3)*4$ is a) 2 b) 3 c) 4 d) 5	1
13.	(i) If $y = x^n$, then the value of $\frac{dy}{dx}$ is a) nx^{n+1} b) nx^{n-1} c) $\frac{x^{n+1}}{n+1}$ d) $\frac{x^{n-1}}{n-1}$ OR (ii) $\int \sec^2 mx \, dx$ is equal to a) $\frac{\sec x}{m} + C$ b) $\sec mx \tan mx + c$	1

	c) $\frac{\tan x}{m} + C$ d) $\frac{\tan mx}{m} + C$	
14.	If $\sin y = x \sin(a + y)$, then $\frac{dy}{dx}$ is a) $\frac{\sin a}{\sin^2(a+y)}$ b) $\frac{\sin^2(a+y)}{\sin a}$ c) $\sin a \sin^2(a + y)$ d) $\frac{\sin a}{\sin^2(a-y)}$	1
15.	(i) Which of the following is a vector quality? a) Mass b) Force c) Time d) Length OR (ii) Which of the following lies in ox'yz octant? a) (4, 2,5) b) (4, -2, 5) c) (-4, 2, 5) d) (4, 2, -5)	1
16.	(i) The distance between the points (3,5,-1) and (9,2,-4) is a) 7 units b) 8 units c) $2\sqrt{6}$ units d) $3\sqrt{6}$ units OR (ii) If the point P divides the line segment joining tow points A(2,-5,2) and B(-3,5,2) internally in the ratio 1:4. Then the coordinators of point P are; a) (-1,3,2) b) (1,-3,2) c) (-1,-3,2) d) (1,-3,2)	1
17.	(i) If OACB is a parallelogram with $\overrightarrow{OC} = \vec{a}$ and $\overrightarrow{AB} = \vec{b}$, then $\overrightarrow{OA}$ is a) $(\vec{a} + \vec{b})$ b) $(\vec{a} - \vec{b})$ c) $\frac{1}{2}(\vec{b} - \vec{a})$	1

	d) $\frac{1}{2}(\vec{a} - \vec{b})$ OR (ii) If $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$ then the unit vector in the direction of $(\vec{a} + \vec{b})$ is a) $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$ b) $\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{k}$ c) $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{k}$ d) $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{k}$	
18.	(i) If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{c} = 3\hat{i} + p\hat{j} + 5\hat{k}$ are coplanar. Then the value of p is a) 4 b) -4 c) 6 d) -6 OR (ii) If a unit vector and $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 15$. Then the value of $\vec{x}$ is a) $-\sqrt{14}$ b) $\sqrt{14}$ c) 4 d) -4	1
19.	If the vectors $3\hat{i} + \lambda\hat{j} + \hat{k}$ and $2\hat{i} - \hat{j} + 8\hat{k}$ are perpendicular to each other then the value of λ is a) 7 b) $\frac{1}{7}$ c) -14 d) 14	1
20.	(i) Converse of the statement; “if n is an odd number, then n^2 is also an odd number is a) If n^2 is not an odd number, then n is also not an odd number b) If n^2 is an odd number, then n is also an odd number c) If n is not an odd number, then n^2 is also not an odd number d) If n is not an odd number, then n^2 is an odd number OR (ii) Contra positive of the statement “If a number is divisible by 6, then it is divisible by 3” is a) If a number is not divisible by 3, then it is not divisible by 6	1

	b) If a number is not divisible by 6, then I is not divisible by 3 c) If a number divisible by 6, then it is not divisible by 3 d) If a number is divisible by 3, then it is divisible by 6					
	Q.No. 21 to 24 are the objective questions of 2 marks each: <i>Some of these questions have 4 sub-parts. You have to do any 2 sub-parts out of 4 sub-parts in such questions.</i>					
21.	Match column –I statement with the right option of column - II	1X2				
	If matrix A is order of 2X3 and matrix B is order of 3X2,					
	<table><tr><th>Column –I</th><th>Column - II</th></tr><tr><td>(i) order of matrix (AB) is (ii) order of matrix (BA) is</td><td>P. 2X3 Q. 2X2 R. 3X2 S. 3X3</td></tr></table>	Column –I	Column - II	(i) order of matrix (AB) is (ii) order of matrix (BA) is	P. 2X3 Q. 2X2 R. 3X2 S. 3X3	
Column –I	Column - II					
(i) order of matrix (AB) is (ii) order of matrix (BA) is	P. 2X3 Q. 2X2 R. 3X2 S. 3X3					
22.	Fill in the blanks: (<i>Attempt any two parts from following questions (i to iv)</i>)	1X2				
(i)	If $y = e^x + c$, then the value of $\frac{dy}{dx}$ is _____					
(ii)	The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^4 - 2\left(\frac{d^2y}{dx^2}\right)^3 - \left(\frac{dy}{dx}\right)^2 + 5 = 0$ is _____					
(iii)	Integrating factor of the differential equation $\sin x \frac{dy}{dx} + y \cos x = 2 \sin^2 x \cos x$ is _____					
(iv)	If $\int_0^a 3x^3 dx = 12$, then the value of 'a' is _____					
23.	Write TRUE for correct statement and FALSE for incorrect statements: (<i>Attempt any two parts from following questions (i to iv)</i>)	1X2				
(i)	A relation of R on a set A defined as $(a,b) \in R \Rightarrow (b,a) \in R$ for all $a,b \in A$ is a symmetric relation.					
(ii)	A relation R is said to be an equivalence relation, if it is reflexive, symmetric but not transitive.					
(iii)	On $A = \{1, 2, 3\}$, a relation R defined as $R = \{(1, 2), (1, 1), (2, 1), (2, 2), (3, 3)\}$ is an equivalence relation					
(iv)	Inverse of bijective functions do not exist.					
24.	Write the negation of each of the following statements:	1X2				
(i)	Sum of 2 and 3 is 6					
(ii)	$a < -7$ or $a > 7$					
	Q.No. 25 to 28 are the objective questions of 4 marks each: <i>These questions have 6 sub-parts. You have to do any 4 sub-parts out of 6 sub-parts in these questions.</i>					
25.	Fill in the blanks: (<i>Attempt any four parts from following questions (i to vi)</i>)	1X4				
(i)	A square matrix A is said to be _____ if $ A = 0$.					
(ii)	A square matrix can be expressed as the sum of a symmetric and a _____ matrix.					
(iii)	If A is a square matrix, then $(A')' =$ _____					
(iv)	The number of all possible matrices of order 2×2 with each entry 0 or 1 is _____					

(v)	Inverse of a square matrix exists only if its determinant is not equal to _____	
(vi)	Co-factor of element (a_{11} i.e. 3) in the matrix $A = \begin{bmatrix} 3 & -5 \\ -4 & 2 \end{bmatrix}$ is _____	
26.	Fill in the blanks: (<i>Attempt any four parts from following questions (i to vi)</i>)	1X4
(i)	If $y = \sec(3x^5)$, then the value of $\frac{dy}{dx}$ is _____	
(ii)	If $y = \cos^2 x$, then the value of $\frac{dy}{dx}$ is _____	
(iii)	The general solutions of the differential equation $\frac{dy}{dx} - 6x = 0$ is given by $y =$ _____	
(iv)	$\frac{d}{dx}(\sin^{-1}x) =$ _____	
(v)	The derivative of the function $\frac{1}{\sqrt{x}}$ is _____	
(vi)	The rate of change of the area of a circle with respect to its radius r at $r = 6$ cm is _____	
27.	Write TRUE for correct statement and FALSE for incorrect statements: (<i>Attempt any four parts from following questions (i to vi)</i>)	1X4
(i)	A function f is said to be decreasing on (a, b) if $f'(x) \geq 0$ for each x in (a, b)	
(ii)	If tangent to a curve $y = f(x)$ at $x = x_0$ is parallel to x -axis, then $\frac{dy}{dx} _{x=x_0} = 0$	
(iii)	For a function $f(x)$: $\int_a^b f(x)dx = -\int_b^a f(x)dx$	
(iv)	$\int_{-a}^a f(x)dx = 0$ if f is an even function.	
(v)	The degree of differential equation $\frac{d^5y}{dx^5} + \log\left(\frac{dy}{dx}\right) = 0$, is not defined.	
(vi)	Differential equation of the family of circles having centre at origin is $x dx + y dy = 0$.	
28.	Carefully study the figure given below and Answer the following:	1X4
		
	(<i>Attempt any four parts from following questions (i to vi)</i>)	
(i)	What is the name commonly given to vectors like vector $\overrightarrow{OP}$?	
(ii)	What is the name given to vectors $\hat{i}$, $\hat{j}$ or $\hat{k}$?	
(iii)	What is the term(s) used for collection of $\hat{i}$, $\hat{j}$ and $\hat{k}$?	
(iv)	In given figure if $ \vec{r} = 10\text{ m}$, $x = 8\text{ m}$ and $y = 6\text{ m}$, Find the value of z ?	

(v)	$\vec{r}$ according to above part (iv) lies in (a) XY Plane (b) YZ plane (c) XZ plane (d) space but not on a XY, YZ, XZ plane.	
(vi)	If $\vec{r}$ makes 30° , 45° and 60° angle with x – axis, y – axis and z – axis, then find the direction cosines of this vector.	
	Q.No. 29 is the objective question of 6 marks: <i>This question has 9 sub-parts. You have to do any 6 sub-parts out of 9 sub-parts in this question.</i>	1X6
29.	Read the passage and answer the questions that follow it. (i to ix)	
	In calculus, optimization problems involve finding the maximum or minimum values of a function, typically within a specified domain. These problems are commonly encountered in various fields, such as economics, physics, and engineering. One fundamental technique for solving optimization problems is to use derivatives. To find the maximum or minimum of a function $f(x)$, we first locate its critical points by setting $f'(x)=0$ and then analyze the behavior of the function around these points using the second derivative test. If $f''(x)>0$ at a critical point, it indicates a local minimum, whereas if $f''(x)<0$, it signifies a local maximum. Integrals, on the other hand, represent the accumulation of quantities over an interval. They are used to find areas under curves and solve problems related to accumulation, such as determining total distance traveled or total accumulated quantities in processes.	
	<i>(Attempt any 6 parts from following questions (i to ix))</i>	
	Consider a function $f(x) = 2x^3 - 3x^2 - 12x + 8$	
(i)	How many critical points does the function have? (a) 0 (b) 1 (c) 2 (d) 3	
(ii)	Which statement about critical points of the function $f(x)$ is correct? (a) All the critical points of $f(x)$ are positive. (b) All the critical points of $f(x)$ are negative. (c) Some critical points of $f(x)$ are positive while others are negative. (d) None of above as function $f(x)$ does not have any critical points.	
(iii)	At positive value of critical point of $f(x)$, function $f(x)$ has	
	(a) Local maximum (b) Local minimum (c) Neither maximum nor minimum (d) None of above as $f(x)$ does not have positive critical points.	
(iv)	At negative value of critical point of $f(x)$, function $f(x)$ has	
	(a) Local maximum (b) Local minimum (c) Neither maximum nor minimum (d) None of above as $f(x)$ does not have negative critical points.	

(v)	Maximum value of $f(x)$ is (a) 10 (b) 15 (c) 20 (d) None of above as $f(x)$ does not have maximum value.	
(vi)	Minimum value of $f(x)$ is (a) -20 (b) -18 (c) -12 (d) None of above as $f(x)$ does not have minimum value.	
(vii)	In the expression $I = \int f(x)dx$, $f(x)$ is termed as (a) Integral (b) Anti – derivative (c) Integrand (d) Derivative	
(viii)	Find the value of $\int f(x)dx$	
(ix)	Find the value of $\int_0^1 f(x)dx$	

SECTION – B		
Question Number	Question	marks
30.	Find the equation of hyperbola with vertices $(\pm 3, 0)$ and the foci $(\pm 5, 0)$. OR Find the equation of the parabola, whose focus is the point $(2, 3)$ and whose directrix is the line $x - 4y + 3 = 0$.	2
31.	Find the value of the determinant of matrix $A = \begin{bmatrix} -2 & 7 \\ -8 & -6 \end{bmatrix}$.	2
32.	Find the minors of the elements of matrix $A = \begin{bmatrix} -5 & 2 \\ -6 & 8 \end{bmatrix}$. OR If $A = \begin{bmatrix} 3 & 2 & 4 \\ 0 & 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 3 & 2 \\ 5 & 1 & 9 \end{bmatrix}$, Find $(A+B)$ and $(A-B)$	2
33.	Find the value of $\sec[\cos^{-1}(\frac{\sqrt{3}}{2})]$	2

34.	Find $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$ OR Evaluate $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^3 - 8}$.	2
35.	If $y = (x^3 - 3x) \cot x$, find $\frac{dy}{dx}$. OR Differentiate $\frac{e^x}{1 + \sin x}$	2
36.	If $A = \pi r^2$, find $\frac{dA}{dr}$ for $r = 2$.	2
37.	Reduce the equation of the plane $4x - 5y + 6z - 120 = 0$ to the intercept form. Find its intercepts on the co-ordinate axes.	2
38.	Find the coordinates of the point which divides the line segment joining the points $(2, 4, 3)$ and $(-4, 5, -6)$ internally in the ratio $2 : 1$.	2
39.	Find the eccentricity, coordinates of the foci and the length of the axes of the ellipse $3x^2 + 4y^2 = 12$.	4
40.	Using elementary transformations, find the inverse of the matrix $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ OR Without expanding the determinants, prove that $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$	4
41.	Prove that : $\tan^{-1} \frac{27}{11} - \tan^{-1} \frac{3}{5} = \cos^{-1} \frac{4}{5}$	4
42.	Find the value of $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ OR Evaluate: $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$	4

43.	Find the equation of the plane passing through the points $(-1, 2, 3)$ and $(2, -3, 4)$ and which is perpendicular to the plane $3x + y - z + 5 = 0$.	4
44.	In a small scale industry a manufacturer produces two types of book cases. The first type requires 3 hours on machine P and 2 hours on machine Q for completion, whereas second type requires 3 hours on machine P and 3 hours on machine Q. The machine P runs at the most 18 hours while the machine Q for at the most 14 hours per day. There is a profit of Rs. 30 on each book case of the first type and Rs. 40 on each book case of second type. How many book cases of each type to be produced for the maximum profit? OR In a village, farmer has 50 hectare of land to grow two crops A and B. The profit from crops A and B per hectare are estimated as Rs. 10,000 and Rs. 9,000 respectively. To control weeds, a liquid herbicide has to be used for crops A and B at rates of 20 litres and 10 litres per hectare. Further, no more than 800 litres of herbicide should be used in order to protect fish and wild life using a pond which collects drainage from this land. How much land should be allocated to each crop so as to maximize the total profit?	6
45.	A square metal sheet of side 48cm. has four equal squares removed from the corners and the sides are then turned up so as to form an open box. Determine the side of the square cut so that volume of the box is maximum. OR Find the cone of maximum volume that can be inscribed in a sphere of radius R.	6

Math 311
Marking Scheme
SECTION A

Q. No.	Correct Answer	Marks
1.	(i) (d) OR (ii) (c)	1
2.	(i) (b) OR (ii) (b)	1
3.	(i) (c) OR (ii) (c)	1
4.	(a)	1
5.	(b)	1
6.	(i) (c) OR (ii) (a)	1
7.	(a)	1
8.	(b)	1
9.	(c)	1
10.	(a)	1
11.	(b)	1
12.	(d)	1
13.	(i) (a) OR (ii) (d)	1
14.	(b)	1
15.	(i) (b) OR (ii) (c)	1
16.	(i) (c) OR (ii) (b)	1
17.	(i) (c) OR (ii) (a)	1
18.	(i) (b) OR (ii) (c)	1
19.	(d)	1
20.	(i) (b) OR (ii) (a)	1
21.	(i)-P, (ii) – S	1 X 2
22.	(i) e^x (ii) 4 (iii) $\sin x$ (iv) ± 2	1 X 2

23.	(i) T (ii) F (iii) T (iv) F	1 X 2
24.	(i) Sum of 2 and 3 is not 6 (ii) $a \geq -7$ and $a \leq 7$	1 X 2
25.	(i) singular (ii) skew – symmetric (iii) A (iv) 16 (v) zero (vi) 2	1 X 4
26.	(i) $15x^4 \sec(3x^5) \tan(3x^5)$ (ii) $2\cos x \sin x$ or $\sin 2x$ (iii) $3x^2 + C$ (iv) $\frac{1}{\sqrt{1-x^2}}$ (v) $-\frac{1}{2x^{3/2}}$ (vi) 12π cm/s	1 X 4
27.	(i) F (ii) T (iii) T (iv) F (v) F (vi) T	1 X 4
28.	(i) Position Vector (ii) unit vector (iii) Unit Orthogonal/ mutually perpendicular Vectors / co – initial vectors. (iv) 0 (v) (a) (vi) $\langle \frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}, \frac{1}{2} \rangle$	1 X 4
29.	(i) (a) (ii) (c) (iii) (b) (iv) (a) (v) (b) (vi) (c) (vii) (c) (viii) $\frac{x^4}{2} - x^3 - 6x^2 + 8x + C$ (ix) $\frac{3}{2}$	1 X 6

Q No.	SECTION B Expected Correct Solution	Step marks	Total Marks
30	Here $a = 3$ and $ae = 5$, Hence $e = \frac{5}{3}$ We know that $b^2 = a^2(e^2 - 1)$ $b^2 = 9\left(\frac{25}{9} - 1\right) = 16$ $\therefore$ Equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$ OR Given focus is $S(2,3)$; and the equation of the directrix is $x - 4y + 3 = 0$. By definition $SP = PM^2$ or $SP^2 = PM^2$ $(x - 2)^2 + (y - 3)^2 = \left\{ \frac{x - 4y + 3}{\sqrt{1^2 + 4^2}} \right\}^2$ $16x^2 + y^2 + 8xy - 74x - 78y + 212 = 0$	1/2 1/2 1/2 ½ OR ½ 1 ½	2
31	We have, $A = \begin{bmatrix} -2 & 7 \\ -8 & -6 \end{bmatrix}$ $\therefore A = (-2) \times (-6) - (-8) \times 7$ $= 12 + 56$ $= 68$	1 1	2
32	We have, $\begin{bmatrix} -5 & 2 \\ -6 & 8 \end{bmatrix}$ $\therefore M_{11} = 8, M_{12} = -6$ $M_{21} = 2, M_{22} = -5$ OR Here both A and B are 2×3 matrices $A + B = \begin{bmatrix} 3+7 & 2+3 & 4+2 \\ 0+5 & 5+1 & 3+9 \end{bmatrix} = \begin{bmatrix} 10 & 5 & 6 \\ 5 & 6 & 12 \end{bmatrix}$ and $A - B = \begin{bmatrix} 3-7 & 2-3 & 4-2 \\ 0-5 & 5-1 & 3-9 \end{bmatrix} = \begin{bmatrix} -4 & -1 & 2 \\ -5 & 4 & -6 \end{bmatrix}$	½X4 = 2 OR 1 1	2
33	Let $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \theta$ Then $\frac{\sqrt{3}}{2} = \cos \theta$ $\cos \theta = \cos \frac{\pi}{6}$ $\theta = \frac{\pi}{6}$ Hence $\sec[\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)] = \sec \frac{\pi}{6} = \frac{2}{\sqrt{3}}$	1/2 1/2 1/2 1/2	2
34	$\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$ Put $\pi - x = h$. $h \rightarrow 0$	½	

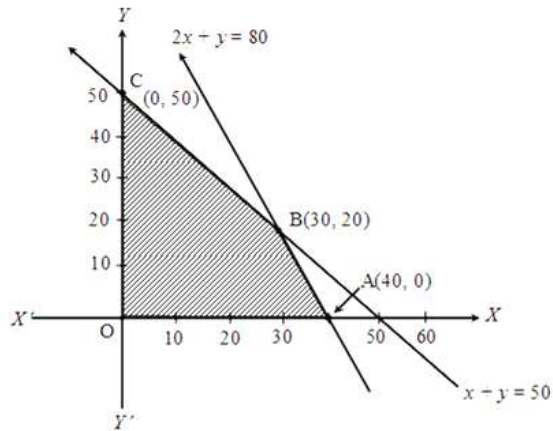
	$\therefore \lim_{h \rightarrow 0} \frac{\sin(\pi - h)}{\pi - (\pi - h)}$ $= \lim_{h \rightarrow 0} \frac{\sinh}{h}$ $= 1$ OR $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^3 - 8} = \lim_{x \rightarrow 2} \left(\frac{x^5 - 32}{x - 2} \right) \div \left(\frac{x^3 - 8}{x - 2} \right)$ $= \lim_{x \rightarrow 2} \left(\frac{x^5 - 2^5}{x - 2} \right) \div \lim_{x \rightarrow 2} \left(\frac{x^3 - 2^3}{x - 2} \right) = 5(2)^4 \div 3(2)^2 = \frac{20}{3} \quad \left[\text{As } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1} \right]$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ OR 1 1	2
35	We have, $y = (x^3 - 3x) \cot x$ $\therefore \frac{dy}{dx} = (x^3 - 3x) \frac{d}{dx} (\cot x) + \cot x \cdot \frac{d}{dx} (x^3 - 3x)$ $= (x^3 - 3x)(-\operatorname{cosec}^2 x) + \cot x(3x^2 - 3)$ $= 3(x^2 - 1) \cot x - (x^3 - 3x) \operatorname{cosec}^2 x$ OR $\frac{d}{dx} \left(\frac{e^x}{1 + \sin x} \right) = \frac{(1 + \sin x) \frac{d}{dx} (e^x) - e^x \frac{d}{dx} (1 + \sin x)}{(1 + \sin x)^2}$ $= \frac{(1 + \sin x)e^x - e^x \cos x}{(1 + \sin x)^2} = \frac{e^x(1 + \sin x - \cos x)}{(1 + \sin x)^2}$	$\frac{1}{2}$ 1 $\frac{1}{2}$ OR 1 1	2
36	Given $A = \pi r^2$ $\therefore \frac{dA}{dr} = \pi \cdot \frac{d}{dr} (r^2)$ $= \pi(2r)$ $= 2\pi r$ At $r=2$, $\frac{dA}{dr} = 2\pi \times 2$ $= 4\pi$	1 1 1	2
37	Equation of the plane is $4x - 5y + 6z - 120 = 0 \text{ or } 4x - 5y + 6z = 120$ This equation can be written as $\frac{x}{30} - \frac{y}{24} + \frac{z}{20} = 1 \text{ or } \frac{x}{30} + \frac{y}{-24} + \frac{z}{20} = 1$ Intercepts on the co-ordinate axes are 30, -24 and 20 respectively.	$\frac{1}{2}$ 1 $\frac{1}{2}$	2
38	Let A (2, 4, 3) and B (-4, 5, -6) be the two points.	$\frac{1}{2}$	

	Again let P(x, y, z) divides AB in the ratio 2 : 1. $\therefore x = \frac{2 \times -4 + 1 \times 2}{2 + 1} = -2$ $y = \frac{2 \times 5 + 1 \times 4}{2 + 1} = 7$ $z = \frac{2 \times -6 + 1 \times 3}{2 + 1} = -3$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	2
39	Given equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$ where $a^2 = 4$ and $b^2 = 3$ $e^2 = 1 - \frac{b^2}{a^2} = \frac{1}{2}$ Coordinates of the foci are (1,0) and (-1,0). Length of the axes are 2a and 2b i.e. 4 and $2\sqrt{3}$.	1 1 1 1	4
40	We have $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ $A = I A$ $\Rightarrow \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$ $\Rightarrow \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A \quad [R_1 \rightarrow R_1 - R_2]$ $\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} A \quad [R_2 \rightarrow R_2 - R_1]$ $\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} A \quad [R_1 \rightarrow R_1 - 2R_2]$ Hence $A^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$ OR L. H. S. = $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$ $C_1 \rightarrow C_1 + C_2 + C_3$	$\frac{1}{2}$ 1 1 1 $\frac{1}{2}$ OR	4

	$= \begin{vmatrix} 2(a+b+c) & b+c & c+a \\ 2(a+b+c) & c+a & a+b \\ 2(a+b+c) & a+b & b+c \end{vmatrix}$ Take 2 common from C_1 $= 2 \begin{vmatrix} a+b+c & b+c & c+a \\ a+b+c & c+a & a+b \\ a+b+c & a+b & b+c \end{vmatrix}$ $C_1 \rightarrow C_1 - C_2$ $= 2 \begin{vmatrix} a & b+c & c+a \\ b & c+a & a+b \\ c & a+b & b+c \end{vmatrix}$ $C_3 \rightarrow C_3 - C_1$ $= 2 \begin{vmatrix} a & b+c & c \\ b & c+a & a \\ c & a+b & b \end{vmatrix}$ $C_2 \rightarrow C_2 - C_3$ $= 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \text{R.H.S.}$	1	
41	$\text{L.H.S.} = \tan^{-1}\left(\frac{27}{11}\right) - \tan^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left\{\frac{\frac{27}{11} - \frac{3}{5}}{1 + \frac{27}{11} \times \frac{3}{5}}\right\}$ $= \tan^{-1}\left(\frac{102}{136}\right)$ $= \tan^{-1}\left(\frac{3}{4}\right)$ $= \cos^{-1}\left(\frac{4}{5}\right)$	$1\frac{1}{2}$ 1 $\frac{1}{2}$ 1	4
42	$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \dots\dots\dots(i)$ $= \int_0^{\pi/2} \frac{\sqrt{\sin\left(\frac{\pi}{2} - x\right)}}{\sqrt{\sin\left(\frac{\pi}{2} - x\right)} + \sqrt{\cos\left(\frac{\pi}{2} - x\right)}} dx$	1 $\frac{1}{2}$	

	$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \dots\dots\dots (ii)$ (i) + (ii) $\Rightarrow 2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$ $\Rightarrow I = \frac{\pi}{4}$ OR $\int \frac{dx}{\sqrt{2-x^2}} = \sin^{-1} \frac{x}{\sqrt{2}} + c$ So $\int_0^1 \frac{dx}{\sqrt{2-x^2}} = \sin^{-1} \frac{x}{\sqrt{2}} \Big _0^1 = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1}(0)$ $= \frac{\pi}{4} - 0 = \frac{\pi}{4}$	$\frac{1}{2}$ 1 1 OR 1 2 1	4
43	Equations of any plane passing through the point $(-1, 2, 3)$ is $a(x+1) + b(y-2) + c(z-3) = 0 \dots\dots\dots (i)$ since the point $(2, -3, 4)$ lies on the plane (i) so $3a - 5b + c = 0 \dots\dots\dots (ii)$ Plane (i) is perpendicular to the plane $3x + y - z + 5 = 0$ hence $3a + b - c = 0 \dots\dots\dots (iii)$ By simplification we get $\frac{a}{4} = \frac{b}{6} = \frac{c}{18} \text{ or } \frac{a}{2} = \frac{b}{3} = \frac{c}{9}$ Required equation of the plane is $2x + 3y + 9z = 31$	$1/2$ $1/2$ 1 1 1	4
44	Let x be the number of book cases of the first type and y that of the second type. Maximize $Z = 30x + 40y$ Subject to the constraints $3x + 3y \leq 18, 2x + 3y \leq 14 \text{ and } x \geq 0, y \geq 0$	$\frac{1}{2}$ $1\frac{1}{2}$	

	Corner points $O(0,0)$, $A(6,0)$, $C(0,14)$ Z at $O(0,0) = 30(0) + 40(0) = 0$ Z at $A(6,0) = 30(6) + 40(0) = 180$ Z at $C(0,14/3) = 30(0) + 40(14/3) = \frac{560}{3}$ Z at $B(4,2) = 30(4) + 40(2) = 200$ $\therefore$ The maximum profit is ` 200-----at the point $B(4,2)$ OR Let x hectare of land be allocated to crop A and y hectare to crop B. Obviously, $x \geq 0, y \geq 0$ Profit per hectare on crop A = Rs. 10000 Profit per hectare on crop B = Rs. 9000 Therefore, total profit = Rs. $(10000x + 9000y)$ The mathematical formulation of the problem is as follows: Maximize $Z = 10000x + 9000y$ Subject to the constraints: $x + y \leq 50$ (which is constraint related to land) ... (i) $20x + 10y \leq 800$ (which is constraint related to use of herbicide), i.e., $2x + y \leq 80$... (ii) $x \geq 0, y \geq 0$ (non negative constraint) ... (iii) Let us draw the graph of the system of inequalities (i) to (iii). The feasible region OABC is shown (shaded) in the figure. The coordinates of the corner points O, A, B and C are $(0, 0)$, $(40, 0)$, $(30, 20)$ and $(0, 50)$ respectively. Evaluation of the objective function $Z = 10000x + 9000y$ at these vertices to find which one gives the maximum profit is done as follows:	2 $\frac{1}{2}$ 1 $\frac{1}{2}$ OR $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 2	6
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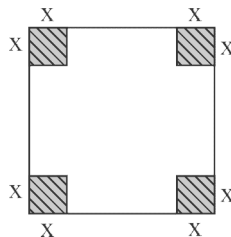
Corner Point	$Z = 10000x + 9000y$
$O(0, 0)$	0
$A(40, 0)$	400000
$B(30, 20)$	480000 ← maximum
$C(0, 50)$	450000

Hence, the farmer will get the maximum profit of Rs. 4,80,000 by allocating 30 hectares for crop A and 20 hectares for crop B.

1½

45

Set the side of each small square be x cm.



∴ Each side of the box to be made is $(48-2x)$ cm, and height x cm.

$V(\text{volume of the box}) = (48-2x)(48-2x)x$

$$\Rightarrow V = (48-2x)^2 \cdot x$$

$$\begin{aligned} \text{Now } \frac{dv}{dx} &= (48-2x)^2 \cdot 1 + 2(48-2x)(-2) \cdot x \\ &= (48-2x)(48-6x) \end{aligned}$$

$$\text{For maxima or minima, } \frac{dv}{dx} = 0$$

$$\therefore (48-2x)(48-6x) = 0$$

$$\Rightarrow x = 24 \text{ or } x = 8$$

$$\Rightarrow 0 < x < 24, \therefore \text{ We have } x = 8$$

$$\begin{aligned} \text{Now } \frac{d^2v}{dx^2} &= 24x - 384 \end{aligned}$$

1

1

1

1

$\frac{d^2V}{dx^2} \text{ at } x=8 = 192 - 384 = -192 < 0$ ∴ V is maximum at x = 8 Hence required side of square to be cut off is 8 cm. OR Let ABC be the cone with radius R $BM = MC = x$ and height $AM = y$ In $\triangle OMB$, $BM^2 + OM^2 = OB^2$ $x^2 + (y - R)^2 = R^2$ $x^2 = 2Ry - y^2$ Volume (V) of the cone $= \frac{1}{3} \pi x^2 y = \frac{\pi}{3} y (2Ry - y^2)$ $= \frac{\pi}{3} (2Ry^2 - y^3)$ $\frac{dV}{dy} = \frac{\pi}{3} (4Ry - 3y^2) = \frac{\pi}{3} y (4R - 3y)$ $\frac{dV}{dy} = 0 \Rightarrow y = \frac{4R}{3}, \text{ (y = 0 is not possible)}$ $\frac{d^2V}{dy^2} = \frac{\pi}{3} (4R - 6y)$ $\frac{d^2V}{dy^2} = \frac{\pi}{3} (4R - 8R) = \frac{-4R\pi}{3} \quad \text{i.e. } \frac{d^2V}{dy^2} \text{ is negative when } y = \frac{4R}{3}$ ∴ V is maximum when $y = \frac{4R}{3}$ Cone has maximum volume when height is $\frac{4R}{3}$ and radius is $\frac{2\sqrt{2}}{3} R$.	1 1 OR 1 1 1 1 1 1 1	6
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